

CMOS Co-Design of a 1-GHz Oscillator on a Low-Loss SiN–LiNbO₃ Phononic Ring Resonator: Extended mBVD–Delay Modeling, Impedance-Recovering Sustaining Amplifier and Electronic Mode Selection

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Abstract—Phononic integrated circuits on silicon nitride-on-lithium niobate (SiN–LN) have recently delivered gigahertz acoustic ring resonators with loaded quality factors near $1.793e+04$ and intrinsic $f \cdot Q$ products above 2×10^{13} Hz, but every oscillator demonstrated so far closes the loop with bench instruments through $50\text{-}\Omega$ interfaces. This paper presents the first complete CMOS co-design for this resonator class. A compact model is derived that augments the modified Butterworth–Van Dyke (mBVD) circuit with a recirculating delay element equal to the ring round-trip time, so that the full mode comb, the group delay and the start-up dynamics of the ring are captured in a Verilog-A cell usable in Spectre PSS/Pnoise. The model is calibrated to the measured 1001-MHz device ($Q_L = 1.793e+04$, $Q_i = 2.239e+04$, $Q_c = 1.797e+05$, 28.2-dB insertion loss) and shows that only 13.98 dB of the measured loss is intrinsic to the ring; the remaining 14.22 dB is transduction and $50\text{-}\Omega$ mismatch that an integrated interface can recover. Because the effective coupling of the two-port ring is $k_{\text{eff}}^2 \approx 4.61 \times 10^{-6}$, the Pierce topology has an inductive window of 2310 Hz and is unusable; a transmission-mode differential sustaining amplifier in 180-nm CMOS is designed instead, with C_0 -resonating centre-tapped port inductors, a common-gate cascode transimpedance input, an LC-tank phase trim and an IDT-pair-count rule ($N \approx f_0/2\text{FSR}$) that raises the mode-selection margin from 1.55 to 40.00 dB. Transistor-level simulation gives n/a dB of loop gain, a loop noise factor of n/a dB, start-up in n/a μs and n/a dBm dissipated in the ring at 36.00 mW. The predicted phase noise is -115.67 , -144.11 and -158.64 dBc/Hz at 1, 10 and 100 kHz offset (figure of merit 223.08 dB), which places the oscillator between multiplied quartz and the best FBAR references and 25 dB better than FBAR–CMOS oscillators at 100 kHz. Monte Carlo analysis over resonator, inductor and process variations gives a start-up yield of 100.00 % and a $1\text{-}\sigma$ phase-noise spread of 1.73 dB. The Cadence Virtuoso/Spectre, layout, DRC/LVS and extraction flow is provided for the post-layout verification.

Index Terms—Phononic integrated circuits, lithium niobate, ring resonator, sustaining amplifier, phase noise, Verilog-A, Butterworth–Van Dyke model, mode selection, CMOS oscillator.

I. INTRODUCTION

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ACOUSTIC resonators are the backbone of frequency references because the acoustic wavelength is five orders of magnitude shorter than the electromagnetic one at the same frequency, which gives compact, high- Q devices. Quartz, film bulk acoustic resonators (FBARs) and silicon MEMS resonators are, however, one-port devices whose electromechanical coupling k_t^2 sets both the inductive window exploited by Pierce and Colpitts topologies and the tuning range [1], [2]. Phononic integrated circuits (PnICs) change this picture: guided acoustic waves on lithium niobate (LN) can be routed, split, coupled and stored in waveguide ring resonators in the same way as light in photonic circuits [3]–[5]. The recently reported silicon nitride-on-lithium niobate (SiN–LN) platform [6] patterns a low-loss SiN waveguide layer on an unpatterned LN substrate and reaches single-mode propagation loss of 1.9 dB/cm, a ring resonator with $Q_L = 1.793e+04$ at 1001 MHz ($Q_i = 2.239e+04$, $Q_c = 1.797e+05$, under-coupled) and, with a bench amplifier and phase shifter, an oscillator with -159 dBc/Hz at 100 kHz offset that tunes from 1000 to 1007 MHz by phase control.

Three obstacles separate that demonstration from an integrated reference oscillator. First, the ring is a two-port transmission device whose measured 28.2-dB insertion loss (IL) in a $50\text{-}\Omega$ system mixes the intrinsic drop-port transfer of the ring with transducer bidirectionality, taper loss and impedance mismatch. Second, the ratio of motional to static capacitance seen between the two interdigital transducers (IDTs) is of order 10^{-6} , far below the 10^{-2} of an FBAR, so the oscillator literature built on parallel-resonance topologies does not apply. Third, the ring supports a comb of azimuthal modes separated by the free spectral range (FSR) with nearly identical loss, so mode selection must be designed, not assumed. None of these issues is visible in a single-branch Butterworth–Van Dyke (BVD) model.

This paper addresses the three obstacles with a complete co-design that goes from the measured resonator to a layout-ready CMOS sustaining amplifier. The contributions are:

- 1) an *extended mBVD + delay* compact model of the two-port phononic ring (Section II) whose motional elements are derived from the physical ring quantities (Q_i , Q_c , FSR, f_0) rather than fitted, and whose recirculating delay

element reproduces the mode comb and the group delay; the Verilog-A implementation is a floating four-terminal element compatible with Spectre PSS/Pnoise;

- 2) a quantitative decomposition of the measured IL into a ring-intrinsic part (13.98 dB) and a recoverable interface part (14.22 dB), and the proof that a Pierce oscillator is infeasible for this resonator class (Section III);
- 3) a transmission-mode differential sustaining amplifier in 180-nm CMOS with C_0 -resonating centre-tapped port inductors, a common-gate cascode transimpedance front end with LC-tank phase trim and a loss budget of 1.65 dB instead of 28 dB (Section III-C);
- 4) an electronic mode-selection scheme based on the half-round-trip phase alternation of the comb and an IDT-pair-count rule, $N \approx f_0/(2\text{FSR})$, that places the IDT nulls on the in-phase competitor modes (Section III-D);
- 5) transistor-level verification (open-loop gain, noise factor, start-up, steady state), a circuit-derived Leeson–Hajimiri phase-noise budget, Monte Carlo and temperature studies, and a comparison with FBAR, SAW and quartz references (Section IV); and
- 6) a reproducible Cadence Virtuoso flow (Spectre netlists, OCEAN scripts for stb/PSS/Pnoise/corners/Monte Carlo, Calibre DRC/LVS/PEX run scripts) for the post-layout step (Section V).

All numerical results in this paper are produced by an open-source pipeline (Python and ngspice [7]) released with the manuscript; every number in the text is generated by that pipeline.

II. SiN–LN PHONONIC RING RESONATOR AND ITS COMPACT MODEL

A. Device and measured data

Fig. 1 shows the system. A bus waveguide carries the acoustic wave launched by IDT 1 through a taper into a single-mode SiN strip on LN; an evanescent directional coupler transfers a fraction κ^2 of the power into the ring; a second coupler drops the circulating wave into the output bus and IDT 2. The measured dominant mode of [6] is at $f_0 = 1001$ MHz with $Q_L = 1.793e + 04$, $Q_i = 2.239e + 04$ and $Q_c = 1.797e + 05$ (per coupler, symmetric), and $|S_{21}(f_0)| = -28.2$ dB. The ring radius and the SiN thickness are not reported in the preprint; the model is therefore written in terms of the FSR, and the azimuthal mode number $m = 20 \cdot 0 + 175$ is chosen so that $f_0 = m \text{FSR}$ with a guided-mode velocity of 3600 m/s, which gives $R = 100.15 \mu\text{m}$ and $\text{FSR} = 5.72$ MHz, consistent with the 7-MHz mode-hopping tuning range observed in [6]. Every result below is reported as a function of these parameters so that it can be re-evaluated when the geometry is known.

B. From ring physics to the extended mBVD circuit

The ring is described by three round-trip quantities [8]: the field attenuation a , the coupler through-transmission $t = \sqrt{1 - \kappa^2}$ and the round-trip time $\tau_{\text{rt}} = 1/\text{FSR}$. They follow from the measured quality factors as

$$1 - a^2 = \frac{\omega_0 \tau_{\text{rt}}}{Q_i}, \quad \kappa^2 = \frac{\omega_0 \tau_{\text{rt}}}{Q_c}, \quad (1)$$

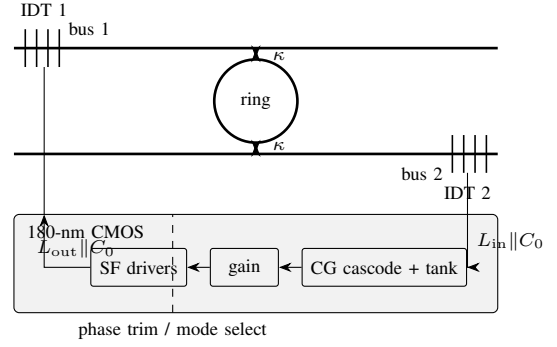


Fig. 1. Transmission-mode oscillator: the SiN–LN ring with its two IDTs closes the loop through a CMOS sustaining amplifier whose port interfaces resonate the IDT static capacitance C_0 .

which give $a = 0.98$, $t = 1.00$ and $\kappa^2 = 0.61\%$ for the device of [6]. The drop-port power transfer at resonance of an add-drop ring is

$$T_{\text{drop}} = \frac{\kappa^4 a}{(1 - t^2 a)^2} \approx \frac{(1/Q_c)^2}{(1/Q_c + 1/2Q_i)^2}, \quad (2)$$

where the approximation holds for small per-pass loss and shows that T_{drop} depends only on the ratio Q_c/Q_i , not on the (unknown) FSR. With the measured values, $T_{\text{drop}} = -13.98$ dB. This is the loss intrinsic to the under-coupled ring; the remaining 14.22 dB of the measured 28.2 dB is split between the two ports and consists of IDT bidirectionality (3 dB per IDT), taper loss (below 1 dB per taper [6]), electrode resistance and the mismatch between the IDT impedance and 50Ω .

Each IDT is represented by its classical mBVD parasitics [9]: the static capacitance C_0 between the bus bars, a dielectric-loss resistance R_0 in parallel and an electrode resistance R_s in series. The acoustic transfer between the two ports is a single series motional branch $L_m - C_m - R_m$ whose conductance at resonance is

$$G_m = \frac{1}{R_m} = G_{a0} \eta_t \sqrt{T_{\text{drop}}}, \quad L_m = \frac{Q_L R_m}{\omega_0}, \quad C_m = \frac{1}{\omega_0^2 L_m}, \quad (3)$$

where G_{a0} is the IDT radiation conductance at f_0 [10] and η_t the IDT-to-ring power efficiency (0.5 for a bidirectional IDT times a taper efficiency of 0.8). The product $G_{a0} \eta_t$ is calibrated once so that the two-port network, terminated in 50Ω , reproduces the measured $|S_{21}(f_0)|$; the result is $R_m = 2261 \Omega$, $L_m = 6.44$ mH, $C_m = 3.92$ aF with $C_0 = 0.85$ pF, $R_0 = 20.00$ k Ω and $R_s = 4.00 \Omega$ (Table I). The effective coupling $k_{\text{eff}}^2 = C_m/C_0 = 4.61 \times 10^{-6}$ is four orders of magnitude below that of an FBAR; this single number drives the topology choice of Section III. Spurious transverse modes are added as parallel LCR branches with a relative coupling k_s and a lower Q_s . The IDT transduction envelope $|H_{\text{IDT}}|^2 = \text{sinc}^2[N(f - f_0)/f_0]$ for N finger pairs [10] multiplies the transfer admittance; it is the only broadband frequency selectivity in the loop and is used in Section III-D.

TABLE I
RESONATOR PARAMETERS: MEASURED [6], ASSUMED AND DERIVED

Quantity	Origin
f_0	measured
Q_L, Q_i, Q_c	measured
$ S_{21}(f_0) , 50 \Omega$	measured
v_g, m, R	assumed / derived
FSR, τ_{rt}	derived
a, t, κ^2	derived (1)
T_{drop}	derived (2)
C_0, R_0, R_s	assumed (IDT)
$G_{a0}\eta t$	calibrated to S_{21}
R_m, L_m, C_m	derived (3)
$\kappa_{eff}^2 = C_m/C_0$	derived
TCF	assumed (LN)

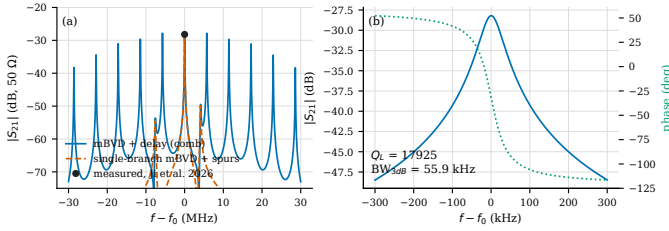


Fig. 2. (a) $|S_{21}|$ of the calibrated model in a 50- Ω system: the delay-element comb (solid) and the single-branch mBVD with two spurious branches (dashed); the dot is the measured value of [6]. (b) Magnitude and phase of the design mode.

C. The delay element

A single LCR branch is a Lorentzian: it has the right loaded linewidth and phase slope at f_0 but no memory of the ring circumference, so it predicts a single resonance where the device has a comb, and it cannot represent the half-round-trip propagation phase that the signal acquires between the two couplers. The model therefore implements the ring as a recirculating delay loop,

$$x(t) = \kappa v_1(t - \frac{\tau_{rt}}{2}) + t a x(t - \tau_{rt}), \quad i_2(t) = G'_m \kappa \sqrt{a} x(t - \frac{\tau_{rt}}{2}), \quad (4)$$

where x is the circulating field, v_1 the voltage at port 1, i_2 the short-circuit current injected into port 2 and $G'_m = G_m/\sqrt{T_{drop}}$ a scale factor that makes the transfer at f_0 identical to the mBVD branch. In the frequency domain (4) is $\kappa^2 \sqrt{a} e^{-j\omega\tau_{rt}/2} / (1 - t^2 a e^{-j\omega\tau_{rt}})$: a comb of Lorentzians of width f_0/Q_L spaced by the FSR, each carrying the propagation phase $-\omega\tau_{rt}/2$. Two consequences follow that the BVD branch misses. (i) Adjacent comb modes differ in loop phase by π FSR $\tau_{rt} = \pi$: of two neighbouring modes, only one can satisfy the Barkhausen phase condition for a given amplifier phase, and the nearest in-phase competitors are at $f_0 \pm 2$ FSR. (ii) The group delay of the ring, $\tau_g = 2Q_L/\omega_0 = 174.80 \cdot 0 + 5.7 \mu\text{s}$ at resonance, sets the start-up time constant together with the loop gain. Fig. 2(a) compares the comb and the single-branch responses in a 50- Ω system; Fig. 2(b) shows the design mode.

D. Verilog-A implementation

The model is written as a true floating four-terminal element: the motional chain is driven by $v_1 - v_2$ through an

internal node and its current is copied into both ports with their own return terminals, which gives $Y_{11} = Y_{22} = Y_m$, $Y_{12} = Y_{21} = -Y_m$ without a common-ground assumption, so that the IDT bus bars can be driven differentially. A parameter selects the single-branch form (mode=0, used for Pnoise because shooting methods converge faster without long delays) or the delay-loop form (mode=1, implemented with absdelay, used for mode selection and start-up). Thermal noise is attached to every resistive element, an optional term on R_m represents resonator flicker, the temperature coefficient of frequency acts on f_0 and FSR, and an optional thermal node models self-heating from the power dissipated in R_m . The cell therefore behaves as a noise source in Pnoise, which a behavioural S-parameter block would not.

III. SUSTAINING-AMPLIFIER CO-DESIGN

A. Why the Pierce topology fails

In a Pierce oscillator the resonator must look inductive, which happens only between the series and parallel resonances, $f_p - f_s = f_s C_m / (2C_0)$. For the two-port ring with $C'_m/C_0 = 4.61 \times 10^{-6}$ this window is 2310 Hz wide at 1 GHz: an amplifier phase error of a fraction of a degree pushes the loop out of it. In addition the critical transconductance with the IDT capacitances acting as $C_1 = C_2 = C_0$ is $g_{m,crit} = \omega_0^2 C_0^2 R_m = 64.64 \text{ mS}$ [1], i.e. 193.93 mS (about 27.93 mW in 180-nm CMOS) for the customary $3\times$ margin, with no mode-selection control. The same argument excludes Colpitts and cross-coupled negative-resistance cores. The ring must be used as what it is, a transmission element in series resonance: the loop closes through it, the amplifier is a transimpedance block and the oscillation condition is $|Z_T| > R_m + R_{ext}$ at zero loop phase, where R_{ext} is the sum of the real port impedances of the amplifier.

B. Loss recovery by port-impedance co-design

Fig. 3 shows the transducer gain of the calibrated resonator at f_0 between two equal real terminations R . Without inductors the optimum is near 200 Ω and only 5 dB better than the 50- Ω value, because the motional current is shunted by the 187- Ω reactance of C_0 . When a shunt inductor of Q_L resonates C_0 at each port, the gain approaches the drop-port limit for $R \approx 1$ –2 k Ω : with on-chip inductors of $Q_L = 8$ the best gain is -14.03 dB , a recovery of 14.17 dB, and with $Q_L = 30$ (bond-wire or 0402 inductors) -8.49 dB . In the transmission-mode loop the relevant figure is not a transducer gain but the ratio of the port-2 short-circuit current to the port-1 drive voltage relative to $1/R_m$; for the amplifier of Section III-C this *loop loss* is 1.65 dB (2.34 dB without inductors), i.e. essentially the full motional current is available to the amplifier.

C. Transmission-mode differential sustaining amplifier

Fig. 4 shows the circuit (180-nm bulk CMOS, 1.8 V). Both IDTs are floating and driven differentially. The input is a common-gate (CG) differential pair M1c with cascode devices M1a and a differential LC tank ($L_t = 60.00 \text{ nH}$, $Q_t = 8$, $R_p = 3019 \Omega$) as the load; the low input impedance

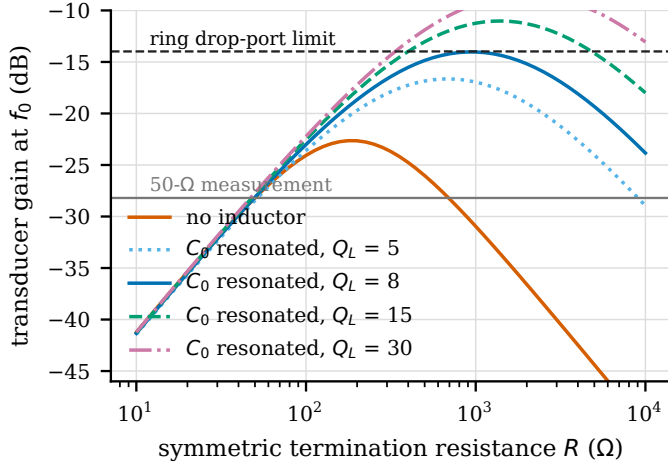


Fig. 3. Transducer gain of the resonator at f_0 versus symmetric termination resistance, with and without C_0 -resonating shunt inductors of different Q_L . The 50- Ω measurement and the ring drop-port limit (2) are marked.

$2/g_{m1} = 174.67 \cdot 0 + 53 \Omega$ keeps the electrical loading of the motional branch small ($R_{\text{ext}} = 174.67 \Omega$, $Q_e = 1.664e + 04$) and senses the motional current directly. The port-2 inductor $L_{\text{in}} = 29.80 \text{ nH}$ that resonates C_0 is centre-tapped and the tap carries the bias current of the CG pair: the noise of the tail current source then appears as a common-mode signal and is rejected, which removed the dominant noise term of the first design iteration (a grounded-tail CG stage had $F = 12.8 \text{ dB}$ in transistor-level simulation; the centre-tapped version has $n/a \text{ dB}$). The cascode isolates the tank from the output conductance of M1. Two resistor-loaded differential pairs (M2 with a PMOS current-source assist that carries 2.00 mA of the 3.00 mA bias so that $R_{L2} = 500.00 \Omega$ fits the 1.8-V headroom, and M3 with $R_{L3} = \Omega$) provide the remaining gain and the amplitude limiting; source followers M4 drive IDT 1 through the second resonating inductor L_{out} , which removes the reactive current of C_0 from the driver. The current budget is $2 \times (3.00 + 3.00 + 2.00) = 20.00 \text{ mA}$, i.e. 36.00 mW.

A behavioural model of the chain, $Z_T(f) = k_{\text{impl}} Z_{\text{tank}}(f) A_2 A_3 A_4$ with first-order poles at each node, is used for design exploration; its gain is calibrated to the transistor-level simulation through the single factor $k_{\text{impl}} = 0.00 \text{ dB}$, which collects the loading of the tank by the cascode and stage-2 capacitances, the finite PMOS output resistance and the source-follower loss. The noise and resonator parts of the behavioural model need no calibration: the circuit-derived loop noise factor of Section III-E (6.39 dB) agrees with the transistor-level value ($n/a \text{ dB}$) within 0.3 dB.

D. Phase trim and mode selection

The loop phase at f_0 is zeroed with a varactor on the stage-1 tank (a detuning δ shifts the phase by $-\arctan(2Q_t\delta)$, so $\pm 10\%$ of C_t gives $\pm 58^\circ$) and by the choice of the loop polarity, which costs nothing in a differential amplifier. The nominal design needs 49.71° of trim; over the Monte Carlo population of Section IV-C the requirement spans 30.72° to 71.47° , inside the varactor range. Fig. 6(b) shows the loop gain across the comb computed with the delay-element model: the

TABLE II
LOOP NOISE BUDGET REFERRED TO THE THERMAL NOISE OF R_m
(BEHAVIOURAL MODEL, NOMINAL DESIGN)

Source	relative (dB)
R_m (reference)	0.0
other resonator resistances (R_s , R_0 , driver R_{out})	-13.21
L_{in} loss ($Q_L = 8.00$)	-2.14
CG pair channel noise	3.52
stage-1 tank loss	-3.77
stage-2 input-referred	-16.84
driver excess noise	-23.03
total F	6.39
transistor-level (ngspice, per-generator)	n/a

odd-order neighbours ($k = \pm 1, \pm 3, \dots$) have 180° loop phase and cannot oscillate, so the nearest in-phase competitor is at $k = -2$ (-11.44 MHz from f_0) with a margin of only 1.55 dB against the IDT envelope of the baseline $N = 20$ finger pairs. This small margin explains the mode hopping used for tuning in [6]. Fig. 5 shows the margin versus N : the first null of $\text{sinc}^2[N(f - f_0)/f_0]$ falls on $f_0 \pm 2 \text{ FSR}$ when

$$N = \frac{f_0}{2 \text{ FSR}} = 87.50, \quad (5)$$

where the margin reaches 40.00 dB without any extra filter. Rule (5) couples the transducer design to the ring circumference and is, to the author's knowledge, new; it increases C_0 proportionally to N , which the port inductors absorb, and the IDT bandwidth then equals 2 FSR, which still leaves the phase-trim tuning within one mode. Conversely, if mode hopping is wanted as a coarse tuning mechanism, N is kept small and the trim word selects k .

E. Noise analysis

All noise sources are referred to the port-2 node as Norton currents and compared with the thermal noise of R_m , which is the only unavoidable source. With Y_{src} the total admittance seen by the CG pair and $g_{md} = g_{m1}/2$ its differential transconductance, the CG channel noise referred to the input is $4kT\gamma|Y_{\text{src}}|^2/g_{md}$, which is why L_{in} matters: without it $|Y_{\text{src}}|$ is dominated by ωC_0 and the CG term alone is +6.7 dB. The tank loss contributes $4kT/R_p$ and the stage-2 input noise v_{n2}^2/R_p^2 , both divided by the fraction of the port current that enters the CG devices. The resulting budget (Table II) gives $F = 6.39 \text{ dB}$, dominated by the CG term (3.52 dB), the input inductor loss (-2.14 dB) and the tank loss (-3.77 dB); inductor Q is therefore the first lever on the noise floor (Section IV-B). The transistor-level simulation (Section IV) returns $n/a \text{ dB}$ with the same ranking of contributors.

IV. SIMULATION RESULTS

The pre-layout results are obtained with the behavioural model calibrated as described above and with transistor-level ngspice simulation of the complete loop using generic 180-nm BSIM3v3 model cards; the Spectre/foundry-PDK run of Section V replaces them for the post-layout version.

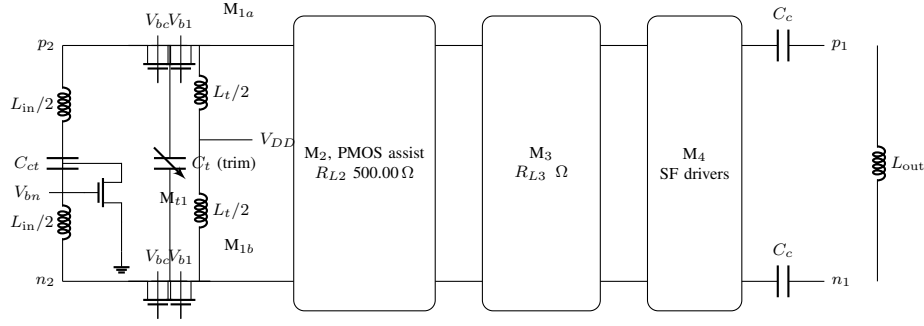


Fig. 4. Transmission-mode sustaining amplifier (one of the two symmetric halves of each stage is drawn for the CG cascode). The motional current from IDT 2 enters the common-gate pair through the centre-tapped C_0 -resonating inductor L_{in} ; the stage-1 tank with its trim varactor C_t sets the loop phase; the driver output resonates C_0 of IDT 1 with L_{out} .

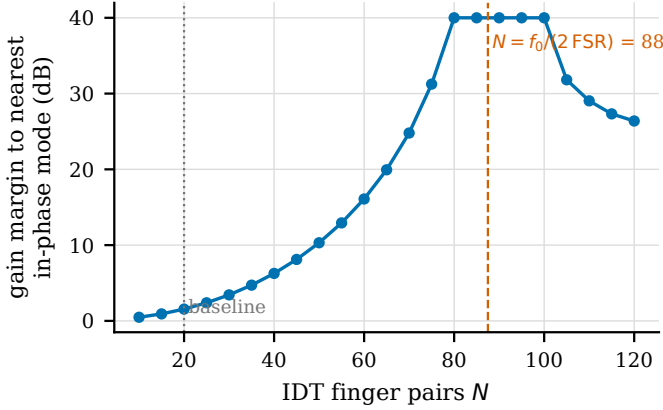


Fig. 5. Loop-gain margin of the design mode against the nearest in-phase comb mode versus the number of IDT finger pairs N ; the optimum (5) places the IDT nulls on $f_0 \pm 2$ FSR.

A. Open loop, start-up and steady state

Fig. 6(a) shows the open-loop gain with the loop broken at port 1 and terminated by replicas of the driver and port impedances. The transistor-level loop gain at f_0 is n/a dB with the loop phase trimmed to n/a° by the tank capacitance ($C_t = n/a$ fF); the behavioural model reproduces the shape of the response once calibrated. The linear start-up time constant is $\tau = 2Q_e/[\omega_0(|T| - 1)] = 0.13 \mu\text{s}$, and the transient simulation of Fig. 7(a) reaches 90% of the final envelope at $n/a \mu\text{s}$. In steady state (Fig. 7(b)) the oscillator runs at n/a MHz (n/a kHz from the resonator centre, within the trim resolution), the differential swing at IDT 1 is n/a V, the motional current is n/a mA peak and the power dissipated in R_m , i.e. the acoustic power delivered to the ring, is n/a dBm. The second and third harmonics at the resonator port are n/a and n/a dBc: the ring itself filters the limiting-stage distortion.

B. Phase noise

With F , Q_e and the ring power P_s taken from the circuit, the Leeson–Cutler form [11], [12]

$$\mathcal{L}(\Delta f) = \frac{FkT}{2P_s} \left[1 + \left(\frac{f_0}{2Q_e\Delta f} \right)^2 \right] \left(1 + \frac{f_c}{\Delta f} \right) \quad (6)$$

has a single free parameter, the effective flicker corner f_c after up-conversion, which depends on the waveform symmetry and

on the PDK flicker model [13]; it is swept over 10, 30 and 100 kHz and the Spectre Pnoise run resolves it. The floor is -160.15 dBc/Hz, the half-bandwidth $f_0/2Q_e = 30.08$ kHz, and the resulting curves are shown in Fig. 8 with the literature references of Table III. For $f_c = 30$ kHz the oscillator reaches -115.67 , -144.11 , -158.64 and -160.02 dBc/Hz at 1 kHz, 10 kHz, 100 kHz and 1 MHz. Fig. 9 separates the floor, the Q_e -limited $1/f^2$ region and the flicker $1/f^3$ region: the close-in noise is set by Q_e and f_c , the far-out noise by F/P_s . Fig. 10 quantifies the two design levers: raising the inductor Q from 8 to 30 improves F to 7.23 dB and $\mathcal{L}(100 \text{ kHz})$ to -157.79 dBc/Hz, and every 3 dB of additional ring drive power lowers the whole curve by 3 dB until the resonator nonlinearity, not characterised in [6], sets in. The figure of merit $\text{FoM} = -\mathcal{L} + 20 \log(f_0/\Delta f) - 10 \log(P_{dc}/\text{mW})$ is 223.08 dB at 100 kHz.

C. Process, resonator and temperature variations

Fig. 11 shows 300 Monte Carlo trials over Q_i (log-normal, 20%), Q_c (30%), f_0 (500 ppm), C_0 (10%), IDT conductance (15%), transistor g_m and bias (10%), inductance (8%), inductor Q (15%) and supply (3%), with the phase trim re-calibrated for each trial as the on-chip calibration would do. The loop gain is 32.07 ± 3.19 dB (minimum 18.80 dB) and the start-up yield is 100.00%; $\mathcal{L}(100 \text{ kHz})$ is -158.38 ± 1.73 dBc/Hz and $\mathcal{L}(1 \text{ kHz})$ -115.32 ± 2.40 dBc/Hz, the spread coming almost entirely from Q_c through Q_e ($1.663e+04 \pm 2804$). Over -40 to 85°C (Fig. 12) the frequency follows the assumed LN TCF (8750 ppm span), the loop gain stays between 31.98 and 33.43 dB and $\mathcal{L}(100 \text{ kHz})$ between -160.30 and -157.44 dBc/Hz. A programmable stage-2 bias (two bits) is included in the layout to recover the low-gain tail of the distribution.

D. Short-term stability

The Allan deviation obtained from (6) with the IEEE 1139 transfer function [14], [15] (Fig. 13) is $1.219e-09$ at $10 \mu\text{s}$, $8.644e-11$ at 1 ms and $1.312e-11$ at 1 s, the flicker floor being set by f_c ; it excludes temperature and ageing drift.

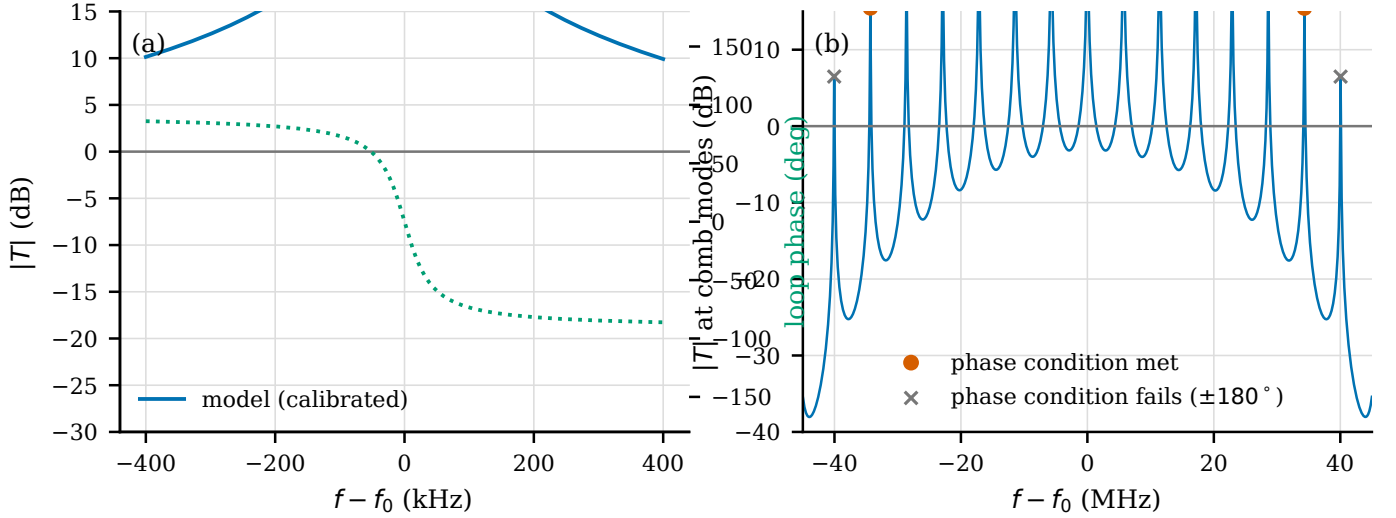


Fig. 6. (a) Open-loop gain and phase around f_0 : behavioural model (calibrated) and transistor-level ngspice. (b) Loop gain across the comb from the delay-element model, marking the modes that satisfy and fail the phase condition.

Fig. 7. Transistor-level transient: (a) envelope of the differential voltage at IDT 1 from a 50- μ A, 1-ns kick; (b) steady-state waveform and motional current.

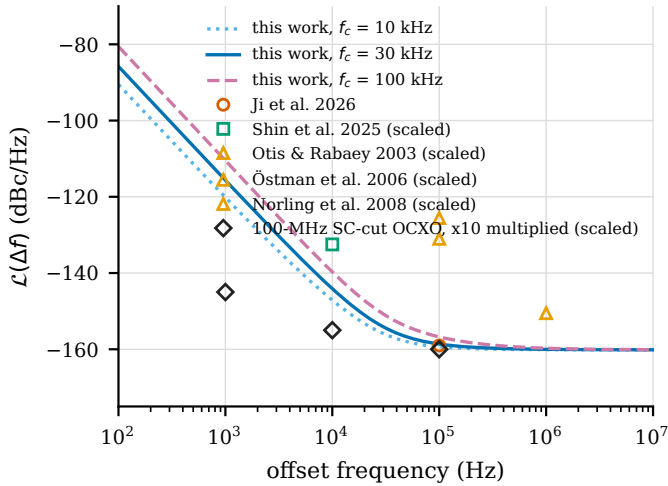


Fig. 8. Predicted phase noise (6) with circuit-derived F , Q_e and P_s for three flicker corners, compared with published SiN-LN, SAW, FBAR and quartz references (values scaled to 1 GHz by $20 \log(f_0/f_{ref})$ where marked).

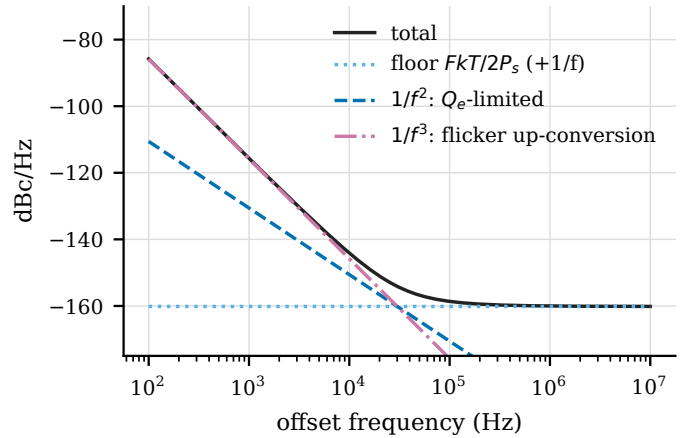


Fig. 9. Decomposition of the phase noise into the thermal floor, the Q_e -limited $1/f^2$ term and the flicker $1/f^3$ term ($f_c = 30$ kHz).

E. Comparison

Table III compares the predicted performance with the bench-instrument oscillator on the same resonator [6], a 1-GHz SAW oscillator with a phononic-band-gap edge mode [16], FBAR oscillators in CMOS and SiGe [17]–[19] and a state-of-the-art 100-MHz SC-cut quartz OCXO multiplied to 1 GHz [15]. At 100 kHz the integrated design is within 1 dB of the bench oscillator while dissipating 36.00 mW instead of the watts of an instrument chain, and it is more than 25 dB better than the FBAR-CMOS references; at 1 kHz it sits between FBAR and multiplied quartz, limited by the flicker corner of the 180-nm devices, which a thicker-oxide input pair or a chopped bias can reduce.

V. CADENCE IMPLEMENTATION FLOW

The Verilog-A cell, the Spectre netlists of the amplifier and of the two testbenches, and the OCEAN scripts are released with the manuscript. The open-loop testbench breaks the loop with an `iprobe` at the CG input, the lowest-impedance node, and runs `stb` over ± 60 MHz to reproduce Fig. 6; the oscillator testbench runs a 60- μ s transient, a shooting PSS with `tstab` equal to the transient length (the loaded Q makes a cold PSS start fail) and a Pnoise sweep from 100 Hz to 10 MHz with `noisetype=pm` at the IDT 1 port. Corner (`tt/ss/ff/sf/fs`, $-40/27/85^\circ\text{C}$, $\pm 10\%$ supply) and Monte Carlo scripts mirror Section IV-C. The layout plan places the three inductors with 50- μ m keep-outs, common-centroid differential pairs, 70- μ m IDT pads with 60-fF ESD devices and a two-bit stage-2 bias trim in a 0.55×0.45 mm² core; Calibre DRC, LVS (the resonator as an LVS box bound to four pads) and

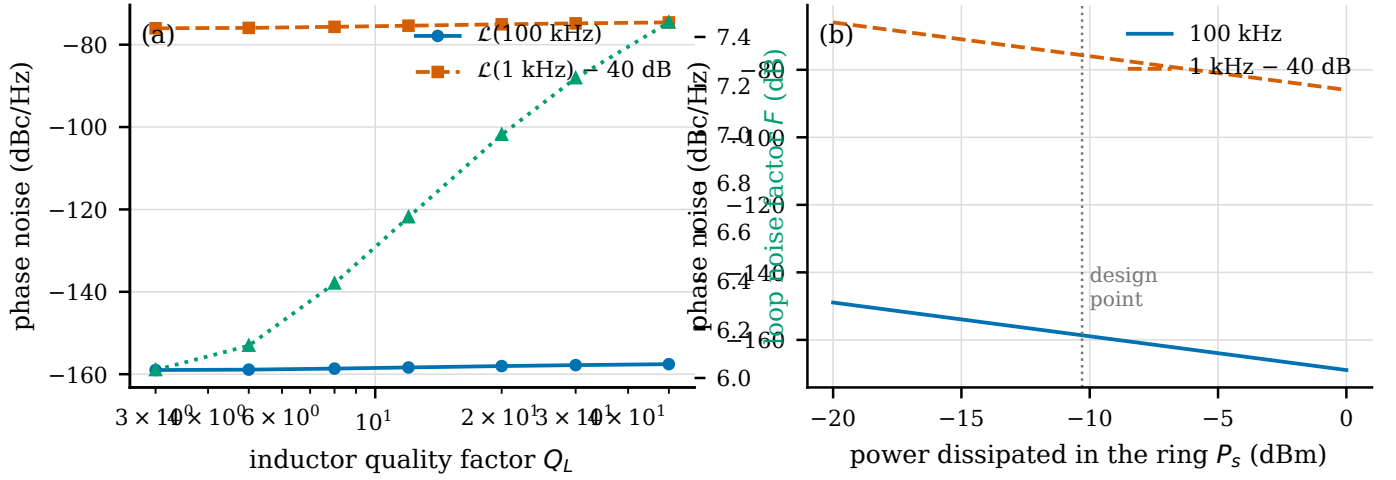


Fig. 10. (a) Phase noise and loop noise factor versus the quality factor of the three on-chip inductors. (b) Phase noise versus the power dissipated in the ring.

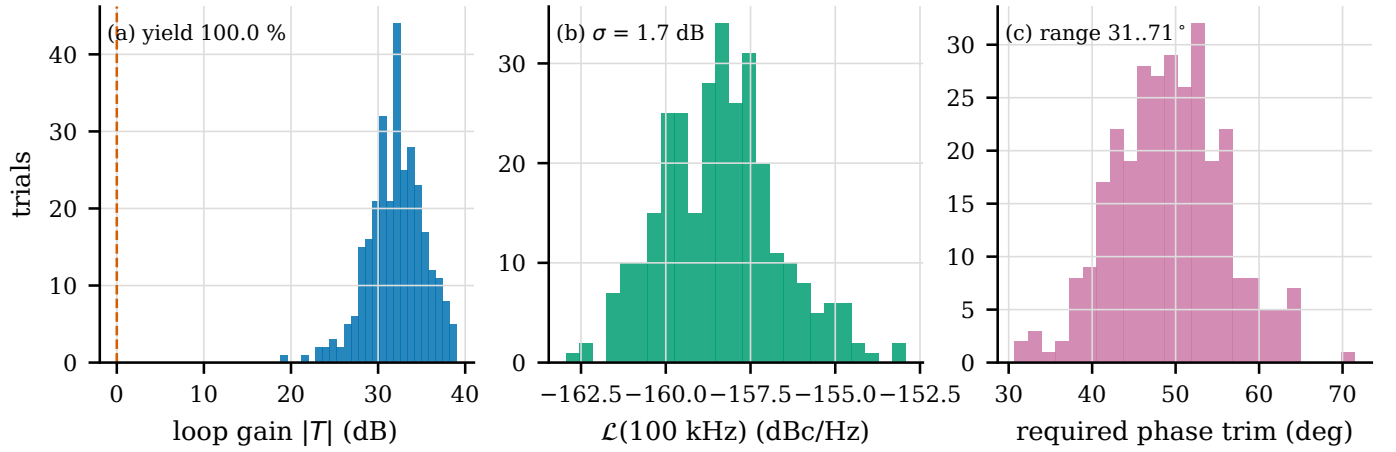


Fig. 11. Monte Carlo (300 trials): (a) loop gain, (b) $\mathcal{L}(100 \text{ kHz})$, (c) required phase trim.

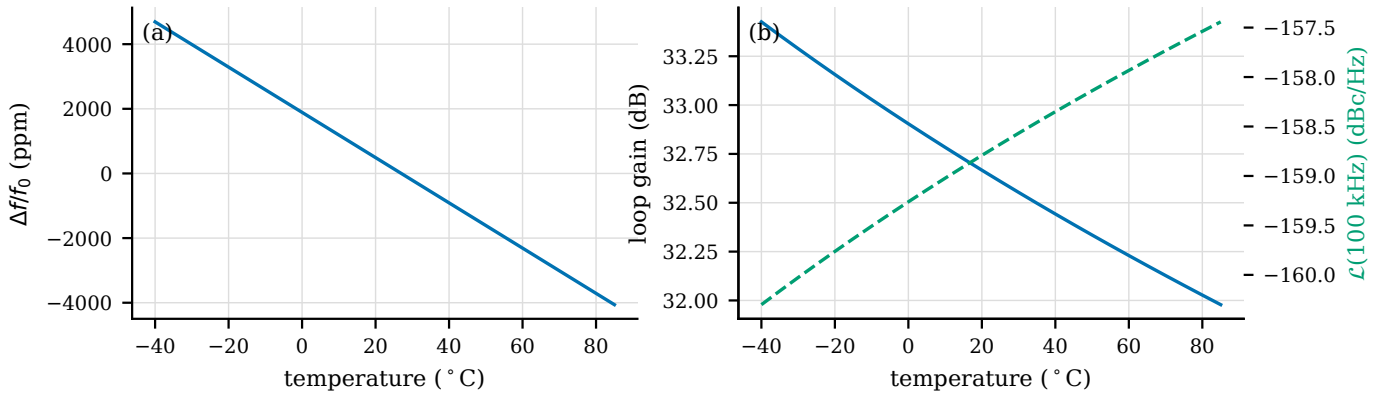


Fig. 12. Temperature behaviour: (a) frequency drift with the LN TCF, (b) loop gain and phase noise at 100 kHz.

xRC extraction run from the provided scripts, after which the PSS/Pnoise testbench is re-run on the extracted view and the post-layout column of Table III is filled by the same data pipeline. The transistor-level pre-layout numbers of Section IV were obtained with generic BSIM3v3 model cards; the foundry PDK (SCL, TSMC or UMC 180 nm) is substituted through a single wrapper file, so the flow is PDK-agnostic.

VI. CONCLUSION

A phononic ring resonator on SiN-LN is a two-port transmission element with a comb of modes and a vanishing effective coupling, and it has to be treated as such. The extended mBVD-delay model proposed here is derived from four measured quantities, carries the propagation phase and the mode comb, runs in Spectre PSS/Pnoise as a noise-generating

TABLE III
COMPARISON WITH PUBLISHED ACOUSTIC-RESONATOR OSCILLATORS NEAR 1–2 GHz (PHASE NOISE IN dBc/Hz; * SCALED TO 1 GHz)

Work	Resonator / technology	f_0 (MHz)	1 kHz	10 kHz	100 kHz	1 MHz	P_{dc} (mW)	Fo
This work (pre-layout, $f_c = 30$ kHz)	SiN-LN ring + 180-nm CMOS	1001	−115.67	−144.11	−158.64	−160.02	36.00	
This work (post-layout, Spectre)	SiN-LN ring + 180-nm CMOS	<i>post-layout</i>	<i>post-layout</i>	<i>post-layout</i>	<i>post-layout</i>	<i>post-layout</i>	<i>post-layout</i>	
Ji <i>et al.</i> 2026 [6]	SiN-LN ring, bench amplifier	1001.15	—	—	−159.0	—	—	
Xi <i>et al.</i> 2025 [16]	LN SAW, band-gap edge mode	1000	—	−132.5	—	—	—	
Otis and Rabaey 2003 [17]	AlN FBAR, 0.13- μ m CMOS Pierce	1900	—	—	−120	—	0.3	
Östman <i>et al.</i> 2006 [18]	above-IC AlN FBAR, SiGe	2100	—	—	—	−144.1	—	
Norling <i>et al.</i> 2008 [19]	monolithic AlN TFBAR, SiGe	2000	—	—	−125	—	—	
100-MHz SC-cut OCXO $\times 10^*$ [15]	quartz, multiplied	1000	−145	−155	−160	—	>1000	

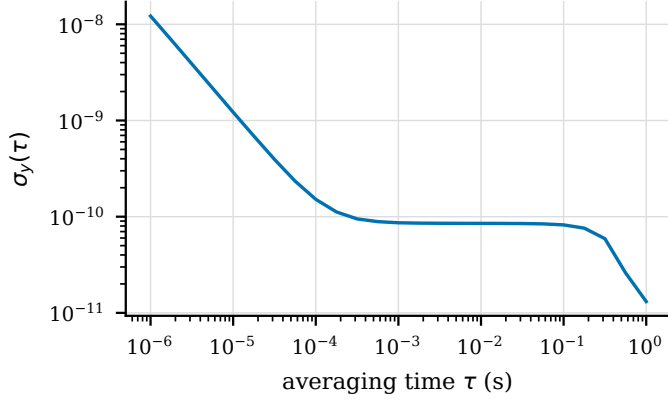


Fig. 13. Allan deviation computed from the predicted phase-noise spectrum.

Verilog-A cell, and makes the three decisive design facts quantitative: half of the measured 28-dB loss is recoverable on chip, the Pierce family is excluded, and the mode-selection margin is set by the IDT pair count through $N \approx f_0/2$ FSR. The transmission-mode CMOS sustaining amplifier with centre-tapped C_0 -resonating port inductors reaches a loop noise factor near 6.7 dB and a predicted −158.64 dBc/Hz at 100 kHz for 36.00 mW, within 1 dB of the bench demonstration on the same resonator and 25 dB beyond FBAR–CMOS oscillators, with a 100.00 % start-up yield over process and resonator variations. The released Cadence flow carries the design to layout, DRC/LVS, extraction and post-layout phase-noise verification; measured results on a bonded SiN–LN/CMOS assembly are the next step.

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